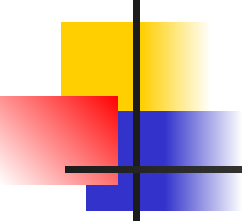
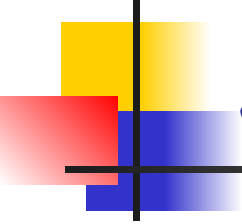


Projectile Motion



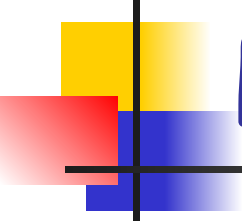
2-Dimensional Motion

- Definition: motion that occurs with both x and y components.
- Example:
 - Playing pool .
 - Throwing a ball to another person.
- Each dimension of the motion can obey different equations of motion.



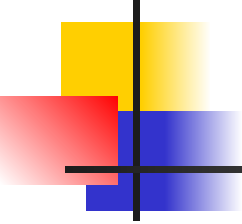
Solving 2-D Problems

- Resolve all vectors into components
 - x-component
 - y-component
- Work the problem as two one-dimensional problems.
 - Each dimension can obey different equations of motion.
- Re-combine the results for the two components at the end of the problem.



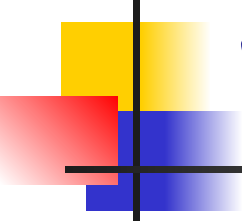
Projectile Motion

- Something is fired, thrown, shot, or hurled near the earth's surface.
- **Horizontal velocity is constant.**
- Vertical velocity is accelerated.
- Air resistance is ignored.



2-Dimensional Projectile

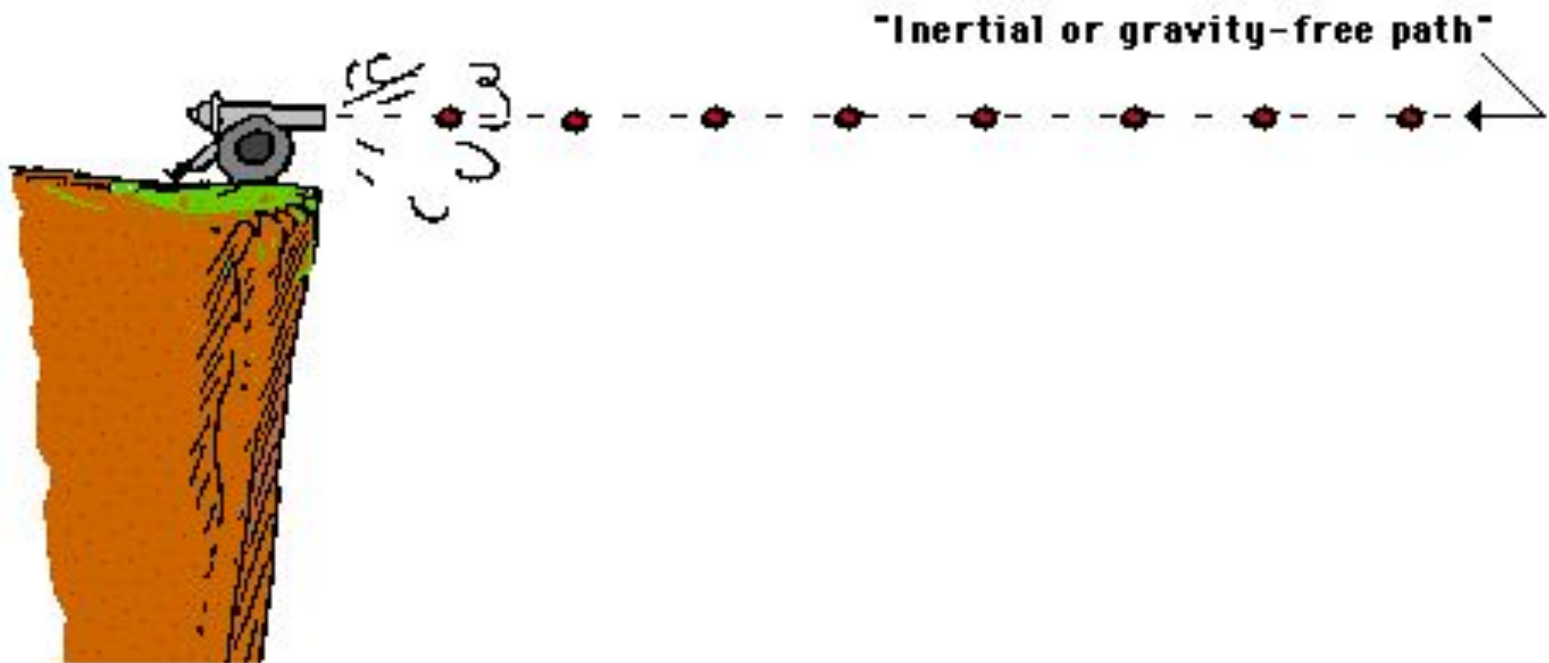
- Definition: A projectile that moves both horizontally and vertically, subject to acceleration by gravity in vertical direction.
- Examples:
 - Throw a softball to someone else.
 - Fire a cannon horizontally off a cliff.
 - **Shoot a monkey with a blowgun.**
- You calculate vertical and horizontal motion.



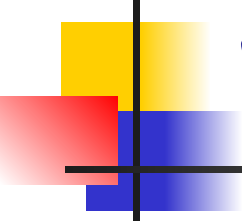
Horizontal Component of Velocity

- Is constant
- Not accelerated
- Not influenced by gravity
- Follows equation:
- $x = V_{o,x} t$

Horizontal Component of Velocity



Without gravity, an object in motion will continue in motion with the same speed and in the same direction.



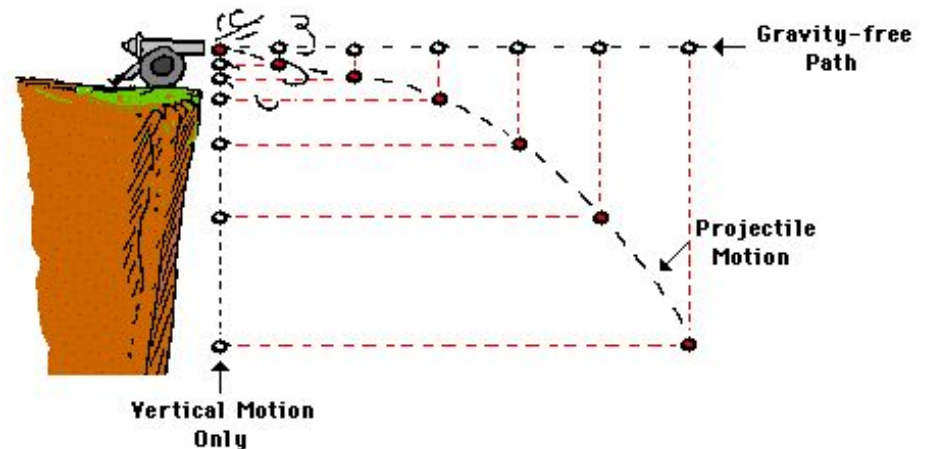
Vertical Component of Velocity

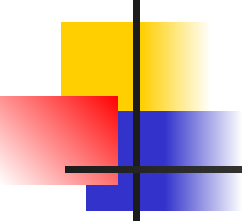
- Undergoes accelerated motion
- Accelerated by gravity (9.8 m/s^2 down)
- $V_y = V_{o,y} - gt$
- $y = y_o + V_{o,y}t - 1/2gt^2$
- $V_y^2 = V_{o,y}^2 - 2g(y - y_o)$

Horizontal and Vertical



Gravity accelerates objects downwards but is unable to effect the horizontal motion of an object.

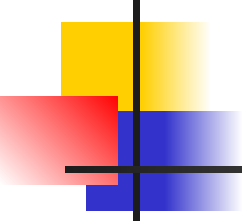




Horizontal and Vertical

**Perpendicular
components of
motion are
independent of
each other.**



- 
-
- In groups of three, determine what are the only factors that affect the time of fall for a horizontal projectile.
 - See if you can come up with an equation that shows this.

Sample Problem

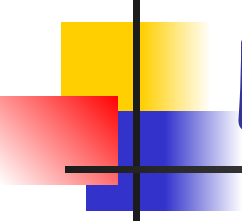
- The Zambezi River flows over Victoria Falls in Africa. The falls are approximately 108 m high. If the river is flowing horizontally at 3.6 m/s just before going over the falls, what is the speed of a drop of water when it hits the bottom? Assume the water droplet is in freefall as it drops.

Sample Problem

- Playing shortstop, you throw a ball horizontally to the second baseman with a speed of 22 m/s . The ball is caught by the second baseman 0.45 s later.
 - a) How far were you from the second baseman?
 - b) What is the distance of the vertical drop?

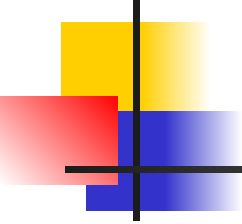
Sample Problem

- An astronaut on the planet Zircon tosses a rock horizontally with a speed of 6.75 m/s . The rock falls a distance of 1.20 m and lands a horizontal distance of 8.95 m from the astronaut. What is the acceleration due to gravity on Zircon?



Launch angle

- Definition: The angle at which a projectile is launched.
- The launch angle determines what the trajectory of the projectile will be.
- Launch angles can range from -90° (throwing something straight down) to $+90^\circ$ (throwing something straight up) and everything in between.



Projectiles launched over level ground

- These projectiles have highly symmetric characteristics of motion.

Trajectory of a 2-D Projectile



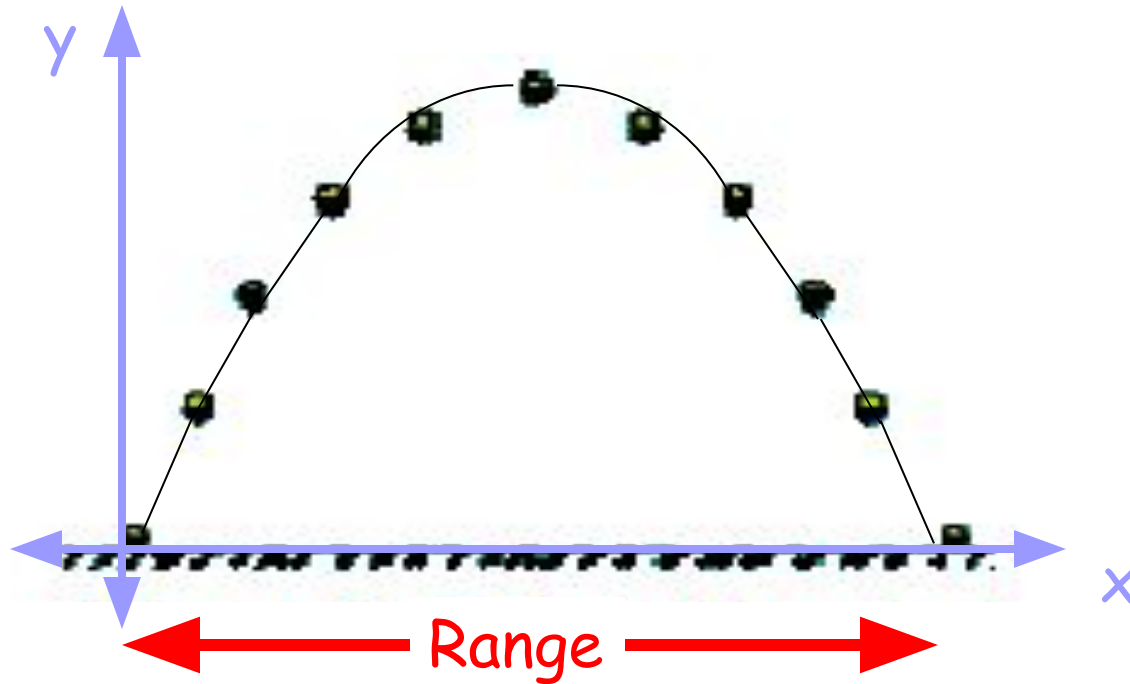
- Definition: The trajectory is the path traveled by any projectile. It is plotted on an x-y graph.

Trajectory of a 2-D Projectile



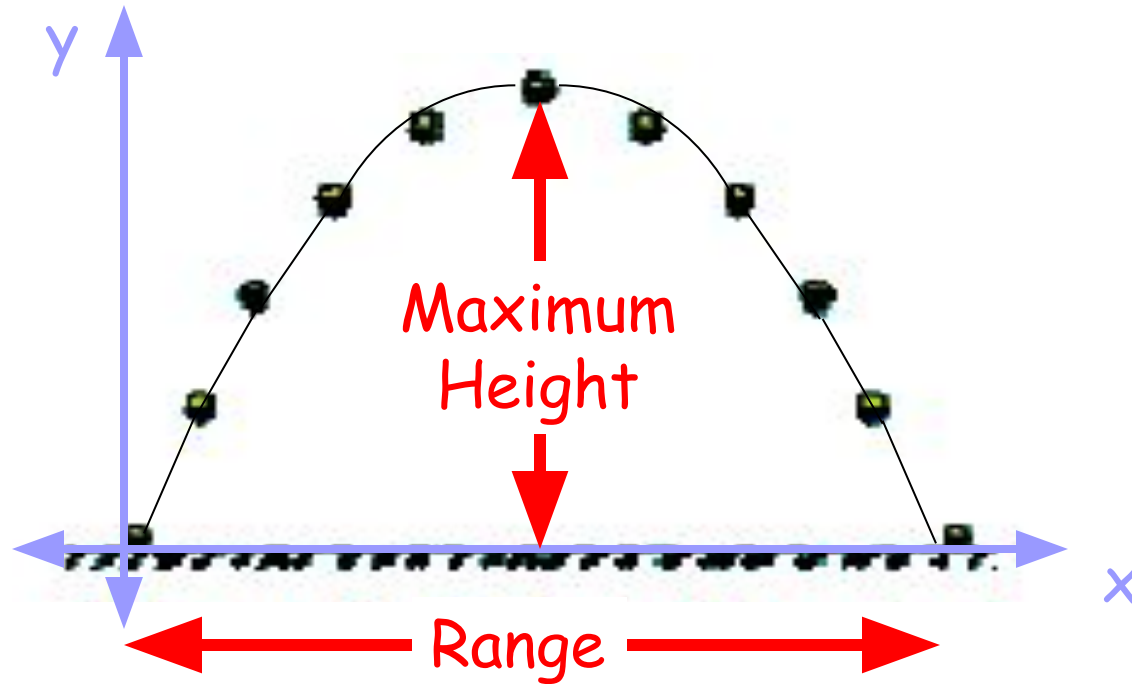
- Mathematically, the path is defined by a parabola.

Range of a 2-D Projectile



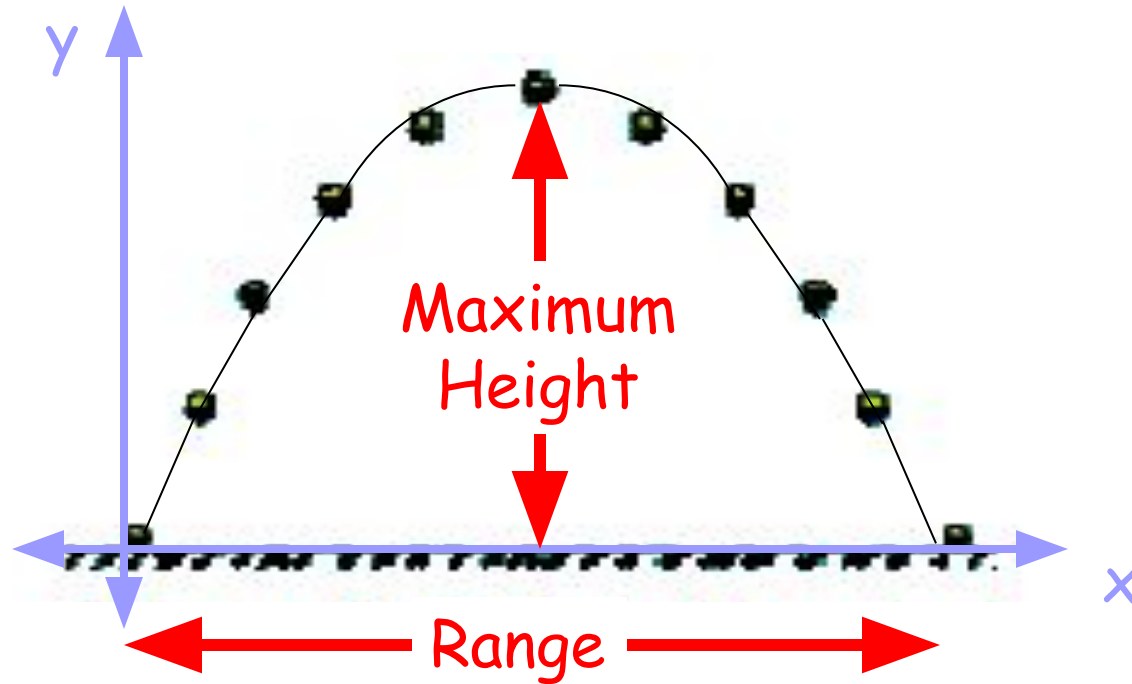
- Definition: The RANGE of the projectile is how far it travels horizontally.

Maximum height of a projectile



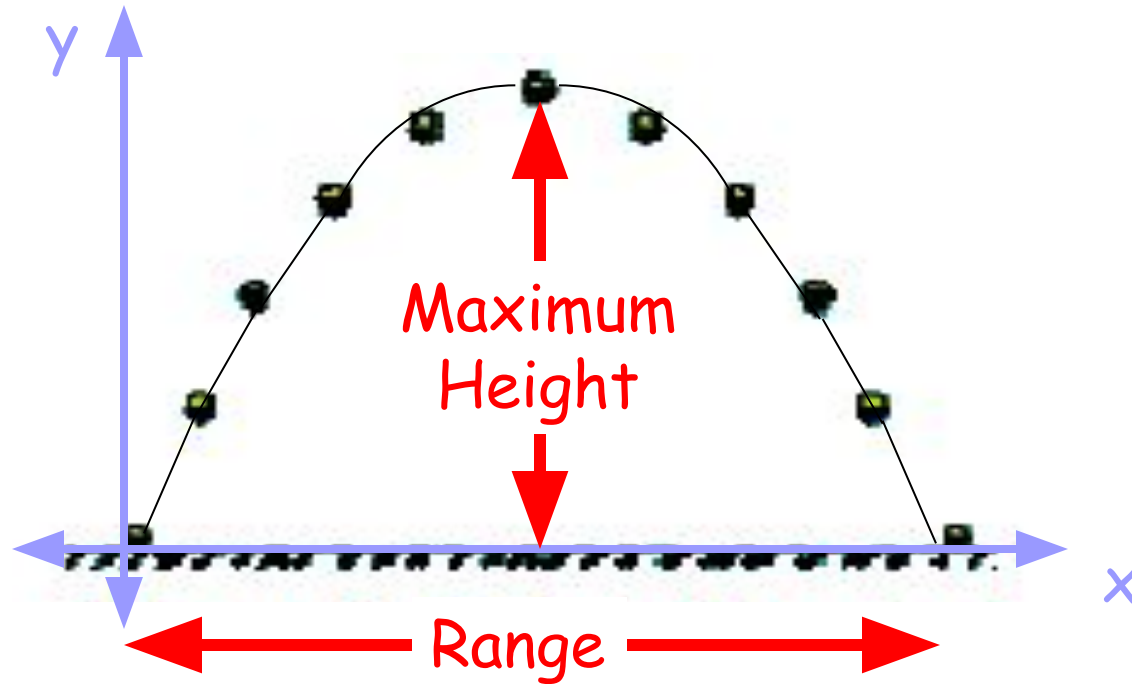
- The **MAXIMUM HEIGHT** of the projectile occurs when it stops moving upward.

Maximum height of a projectile



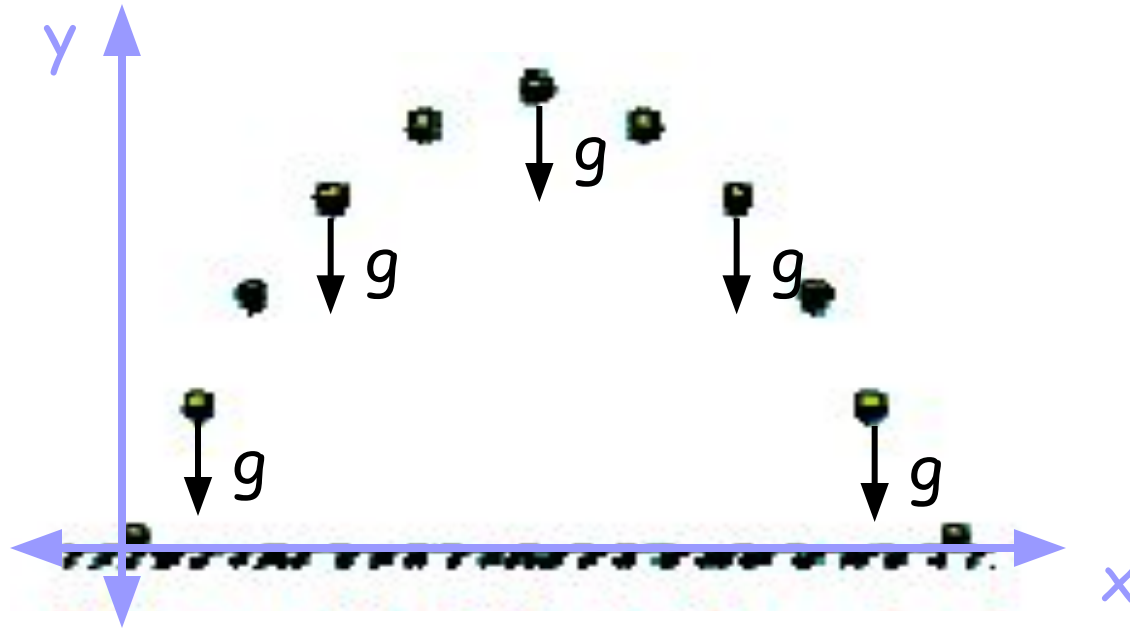
- The vertical velocity component is zero at maximum height.

Maximum height of a projectile



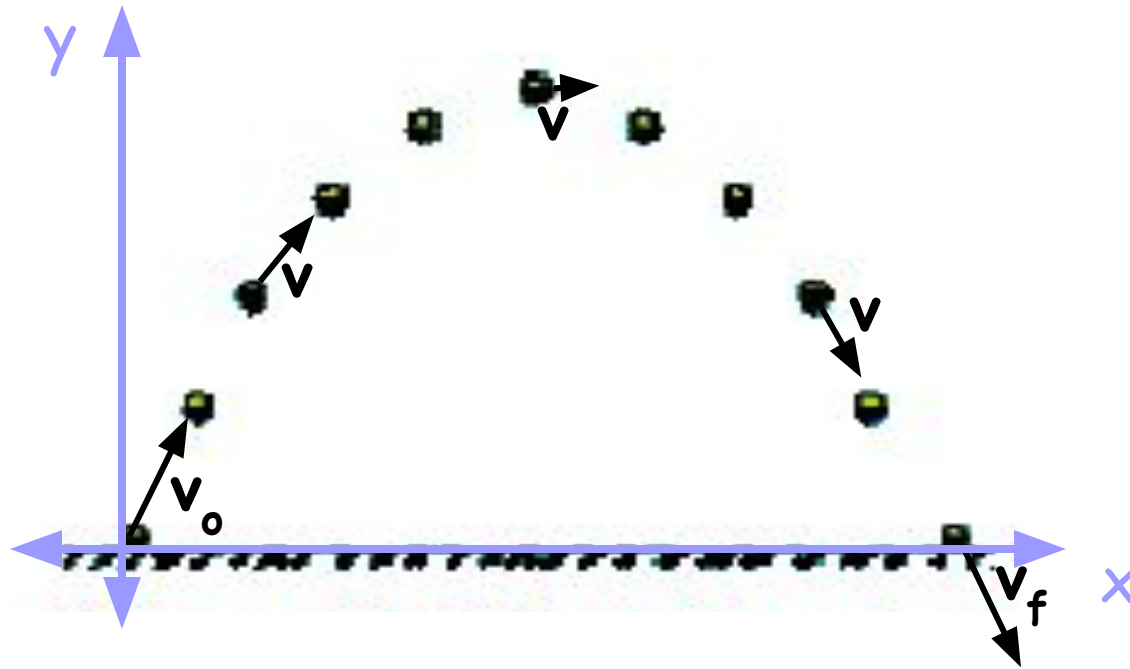
- For a projectile launched over level ground, the maximum height occurs halfway through the flight of the projectile.

Acceleration of a projectile



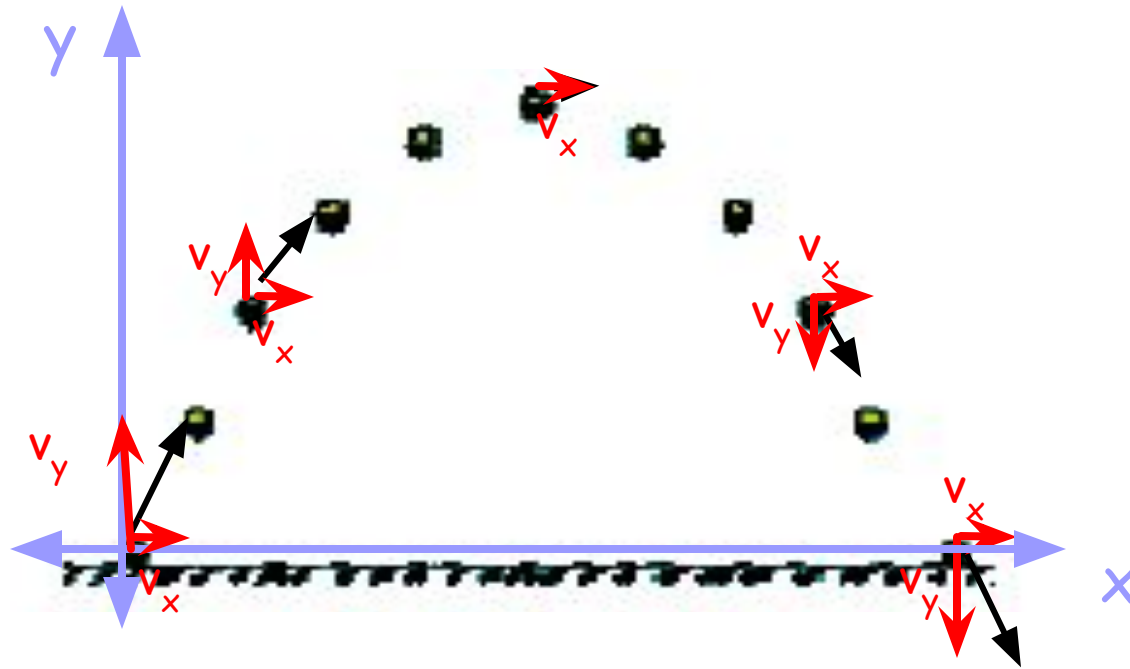
- Acceleration points down at 9.8 m/s^2 for the entire trajectory of all projectiles.

Velocity of a projectile



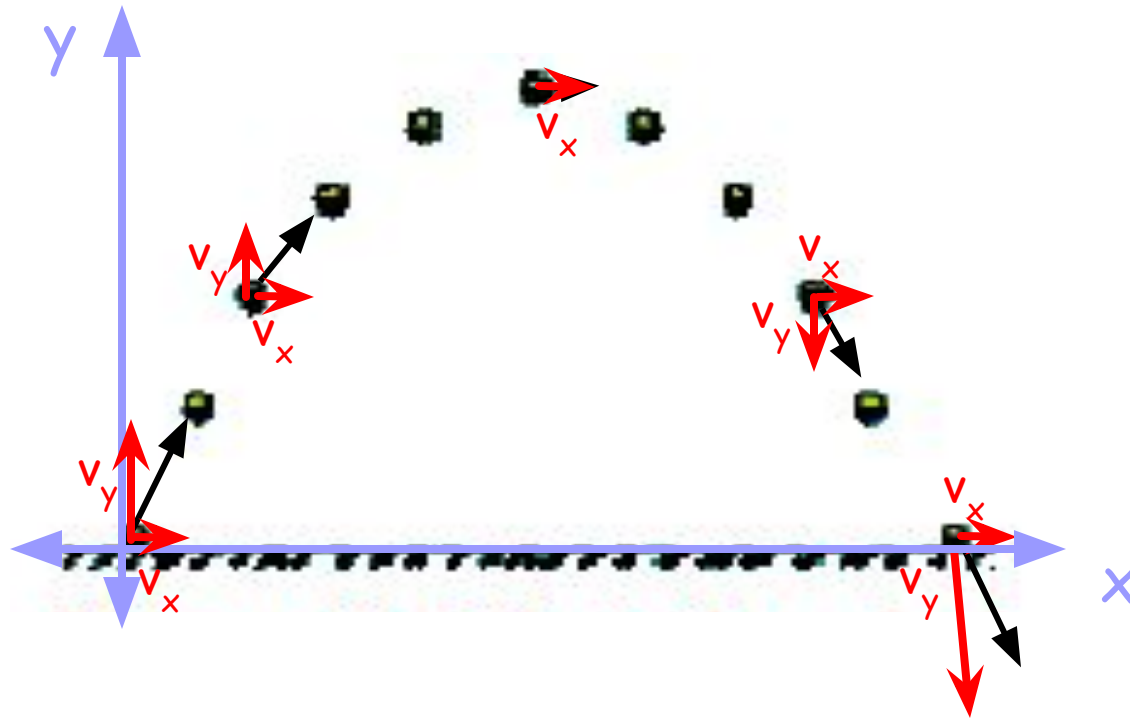
- Velocity is tangent to the path for the entire trajectory.

Velocity of a projectile



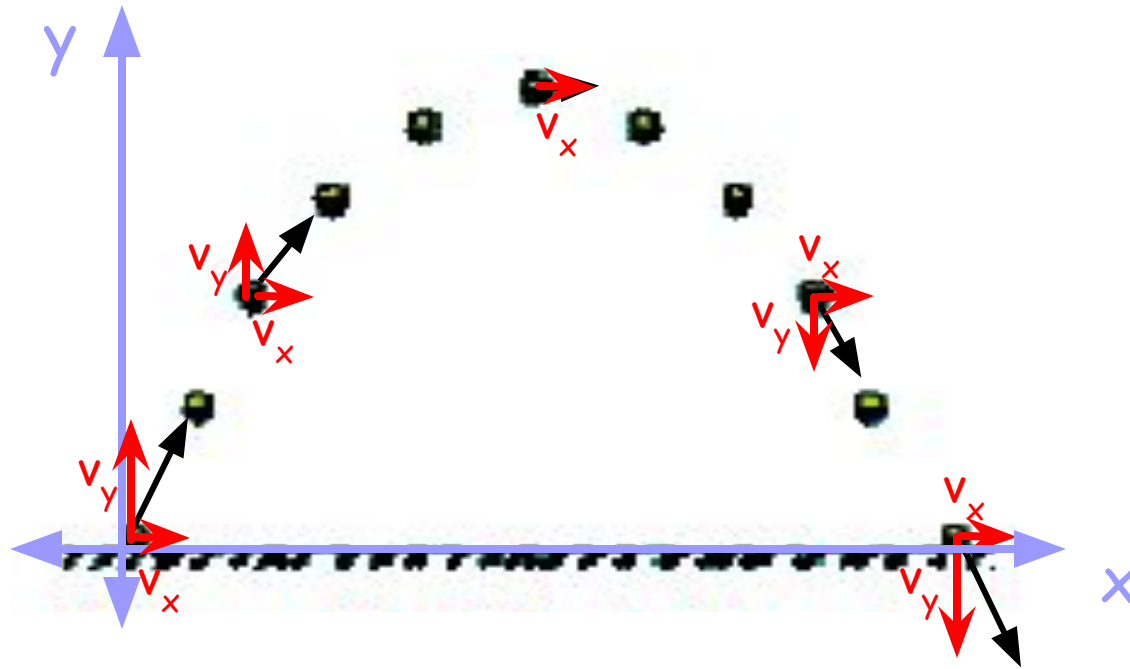
- The velocity can be resolved into components all along its path.

Velocity of a projectile



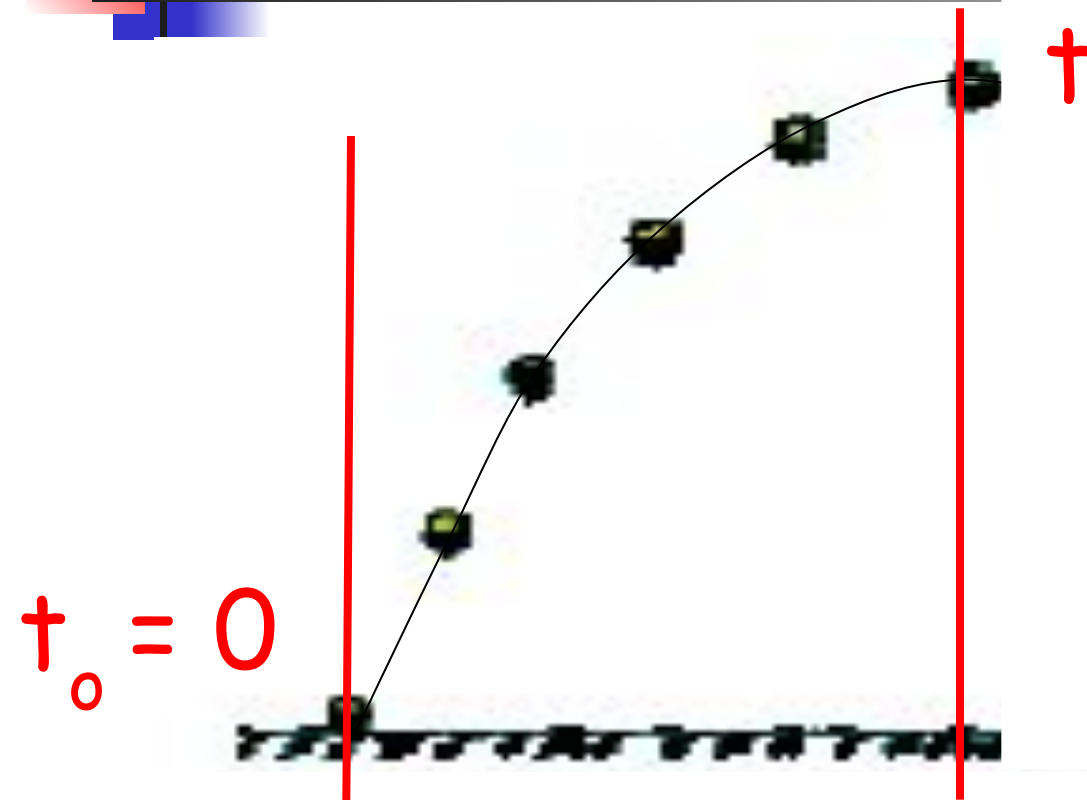
- Notice how the vertical velocity changes while the horizontal velocity remains constant.

Velocity of a projectile



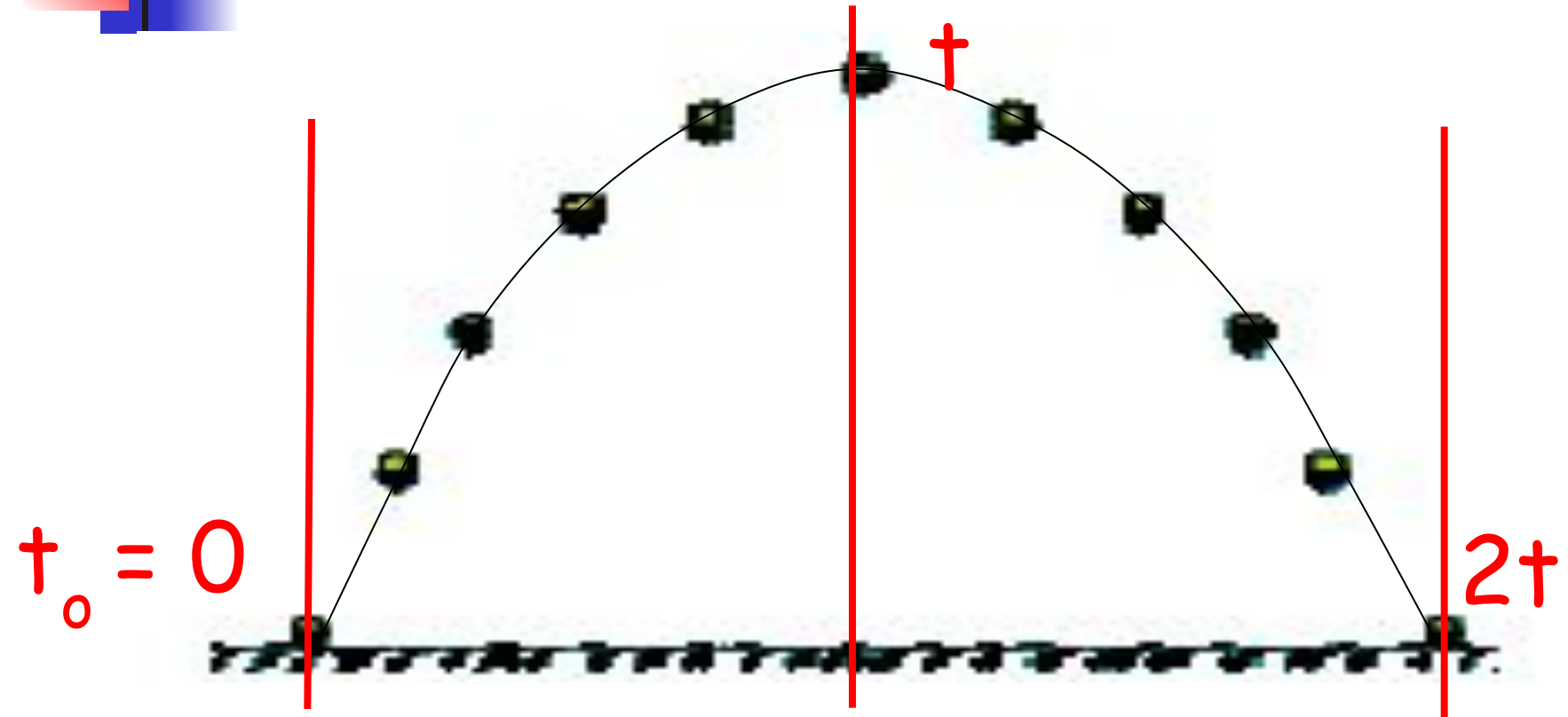
- Maximum speed is attained at the beginning, and again at the end, of the trajectory if the projectile is launched over level ground.

Time of flight for a projectile



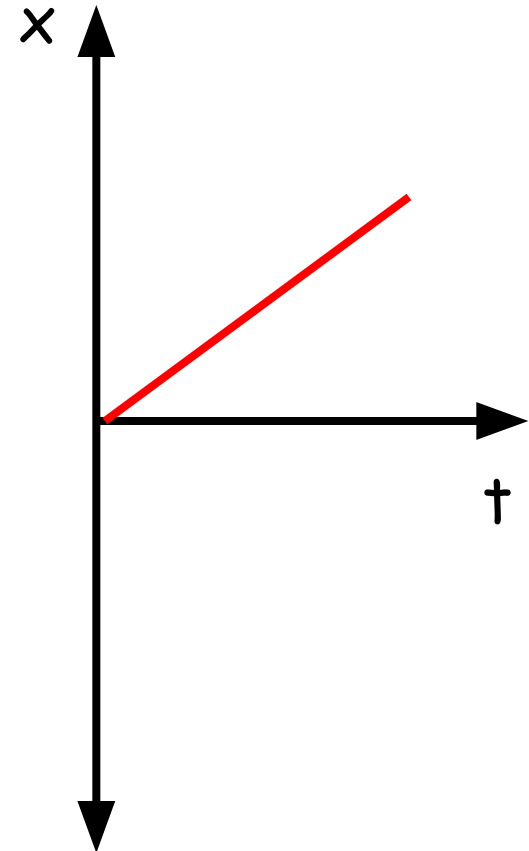
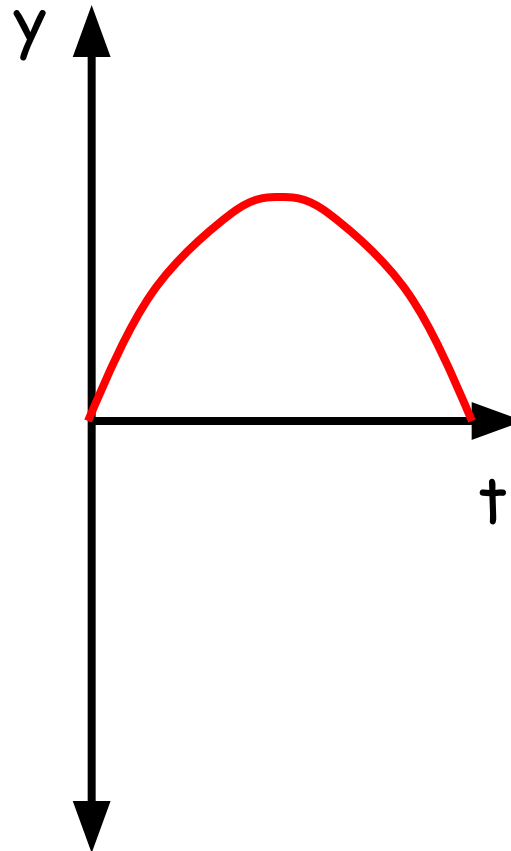
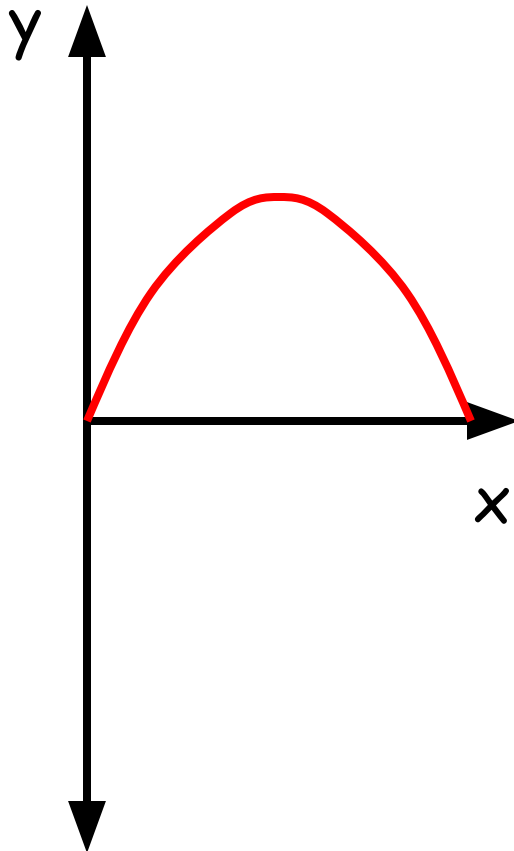
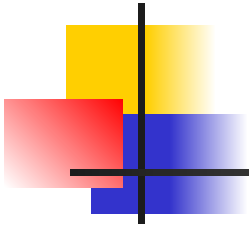
- The projectile spends half its time traveling upward...

Time of flight for a projectile

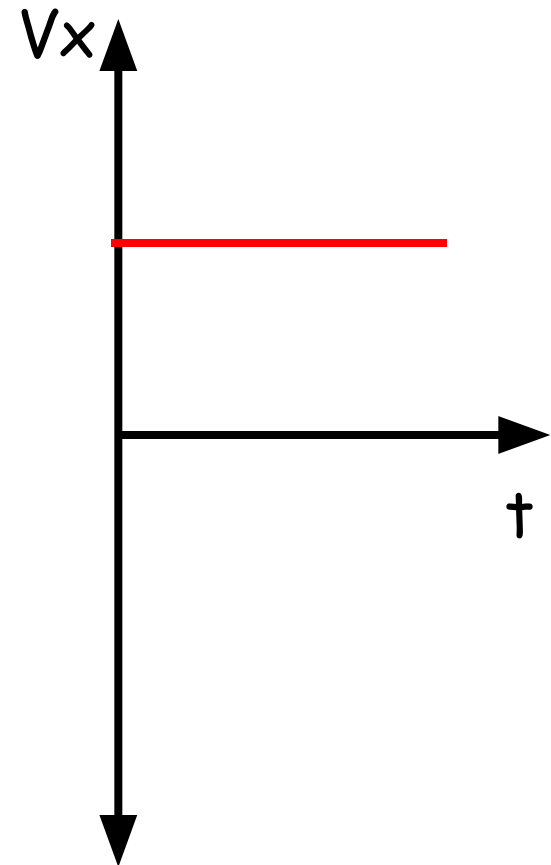
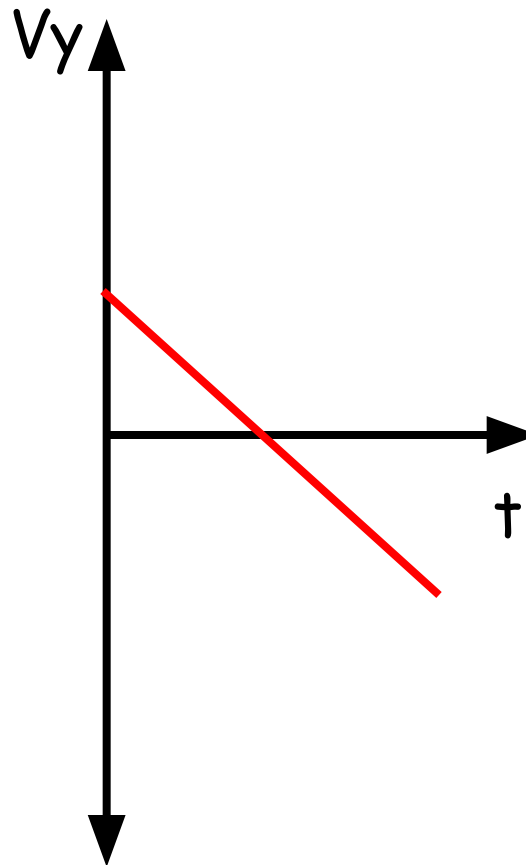


- ... and the other half traveling down.

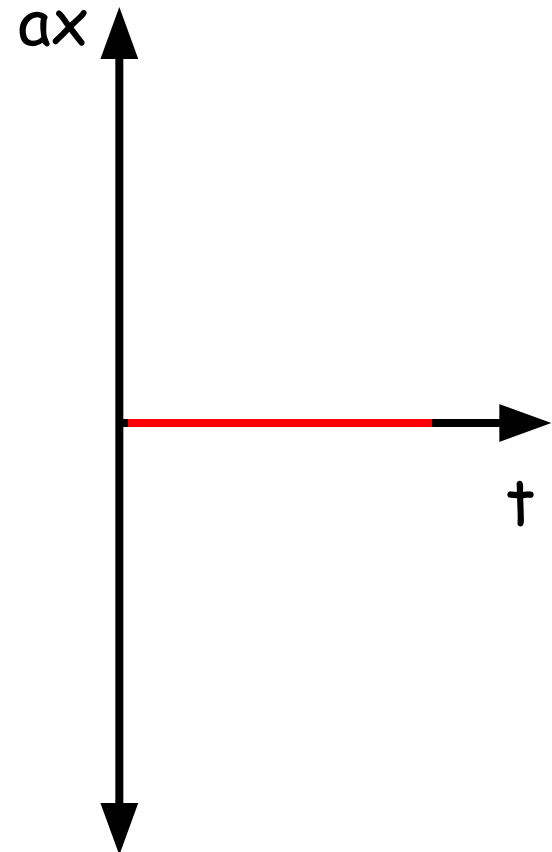
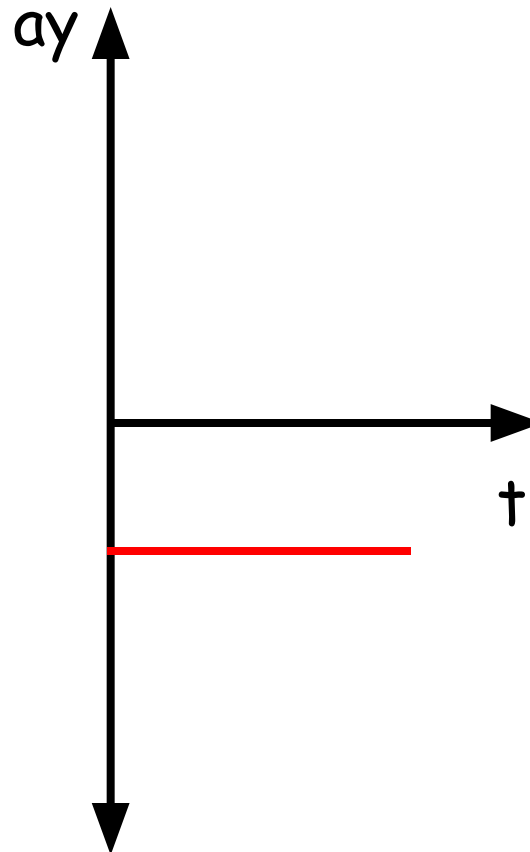
Position graphs for 2-D projectiles



Velocity graphs for 2-D projectiles



Acceleration graphs for 2-D projectiles

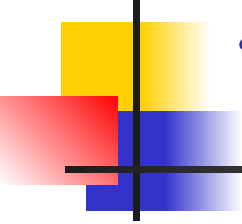


Sample problem

- A soccer ball is kicked with a speed of 9.50 m/s at an angle of 25° above the horizontal. If the ball lands at the same level from which it was kicked, how long was it in the air?

Sample problem

- Snowballs are thrown with a speed of 13 m/s from a roof 7.0 m above the ground. Snowball A is thrown straight downward; snowball B is thrown in a direction 25° above the horizontal. When the snowballs land, is the speed of A greater than, less than, or the same speed of B? Verify your answer by calculation of the landing speed of both snowballs.



The Range Equation

- Derivation is an important part of physics.
- Your book has many more equations than your formula sheet.
- The Range Equation is in your textbook, but not on your formula sheet. You can use it if you can memorize it or derive it!

Sample problem

- A golfer tees off on level ground, giving the ball an initial speed of 42.0 m/s and an initial direction of 35° above the horizontal.
 - a) How far from the golfer does the ball land?

Sample problem

- A golfer tees off on level ground, giving the ball an initial speed of 42.0 m/s and an initial direction of 35° above the horizontal.
- b) The next golfer hits a ball with the same initial speed, but at a greater angle than 45° . The ball travels the same horizontal distance. What was the initial direction of motion?